

Clinical Risk Assessment Report: Patient 20

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 36.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.66x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +5.3%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 1.87 [Top 6%] (+5.3%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.75 [Avg (73th %ile)] (+3.8%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.84 [Top 19%] (+3.1%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 0.02 [Top 11%] (+3.0%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.28 [Top 10%] (+2.5%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.24 [Avg (41th %ile)] (+1.7%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.2 [Bottom 22%] (+1.4%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 66.0 [Avg (59th %ile)] (-2.9%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Machine Learning Feature Attribution (SHAP Decision Path)

